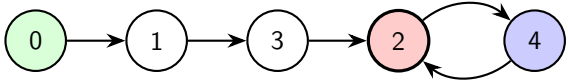
`nums = [1,3,4,2,2]`: long tail ($a = 3$), tiny cycle ($c = 2$)



step	0	1	2	3	4
slow	0	1	3	2	4
fast	0	3	4	4	4

fast walked $8 = a + b + 2c = 3 + 1 + 2 \cdot 2$ (2 laps, $k = 2$)
slow walked $4 = a + b$

Phase 2: the pointer from the meeting point (4) needs $a = 3$ jumps: $4 \to 2 \to 4 \to 2$.
That is one full extra lap ($k - 1 = 1$) plus $c - b = 1$; it lands on 2 together with the pointer from 0.